

Observe, Expand, Recover: Observability-Guided Control for Query Expansion

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Abstract—LLM query expansion is typically applied unconditionally, despite evidence it can be net-harmful on a measurable fraction of queries. We formulate expansion as a bounded, observable agentic decision: observe telemetry, act, verify before committing. Our central finding: telemetry available without relevance judgments at inference time, recorded *before* an action, is reasonably predictive of retrieval failure (AUC 0.649), but recorded *after* the action, it is markedly weaker at telling whether that action helped (AUC 0.552). Detection is not actionability. Neither a static expansion pipeline nor a closed-loop recovery agent beats a properly-tuned retriever on ordinary queries, despite a retrospective oracle indicating uncaptured headroom (+0.0246 nDCG@10) our verifier largely cannot exploit. A six-fault perturbation analysis shows this gap is per-fault, with distinct, interpretable telemetry signatures. The same ordering replicates on an independent second corpus, BEIR Natural Questions (failure AUC 0.680 vs. action-utility AUC 0.556, paired difference 95% CI excluding zero): knowing that retrieval has failed is substantially easier than knowing which intervention will fix it.

Index Terms—query expansion, information retrieval, agentic systems, AI observability, retrieval-augmented generation, large language models

I. INTRODUCTION

Large language models can generate useful pseudo-documents, alternative phrasings, and disambiguating context for a search query [1], [2], and corpus-grounded generation — conditioning the rewrite on retrieved evidence rather than the query text alone — improves on naive query-only rewriting. Existing work establishes that LLM query expansion *can* help, but leaves open when an operational system should trust an expansion enough to use it, and in practice it is applied unconditionally or not at all — neither provides a reliable decision rule: unconditional rewriting can be net-harmful on a measurable fraction of queries, particularly underspecified or already-well-served ones. Our agent alternates actions with structured observations as in ReAct [3] (Section III-D), with accept, roll-back, replan, and abstain decisions driven by runtime telemetry, not the model’s own self-assessment. We study this distinction explicitly: predicting that retrieval is failing and predicting that a specific intervention will fix it are different tasks, and telemetry useful for the first need not be useful for the second.

A. Research Questions

RQ1 (Detection vs. Actionability). Can qrel-free retrieval and agent-runtime telemetry (a) identify when unexpanded retrieval is already unhealthy, and (b) predict whether a *specific* expansion action will help or harm? How much harder is the second task than the first?

RQ2 (Control). Does an agent that conditions expansion actions on that telemetry outperform an otherwise identical *static*, failure-conditioned pipeline — same retriever, grounding, generator, and

fusion, differing only in whether a closed loop can observe an action’s consequences and recover from a bad one?

RQ3 (Recovery). When retrieval is corrupted by a harmful rewrite or a simulated system-level fault, can observability-driven verification detect the degradation and select a recovery action to restore retrieval quality under a bounded budget — even given RQ1’s finding that its action-utility signal is weak?

B. Contributions

We contribute four findings, each under a frozen validation/test protocol (Section IV-B) so no result is conditioned on the split used to report it. **C1, detection vs. actionability:** qrel-free runtime telemetry predicts retrieval failure substantially better than it predicts intervention utility (AUC 0.649 vs. 0.552). Existing selective-expansion and query-performance-prediction work primarily asks whether retrieval is difficult or whether expansion is likely to help; we separate state diagnosis from post-action verification within the same closed-loop system and show empirically that successful diagnosis does not imply actionable intervention selection. **C2, consequences for agentic control:** a matched static-vs-agent evaluation, plus an oracle and selectivity-frontier analysis, shows substantial control headroom exists (+0.0246 nDCG@10) but is mostly inaccessible to the verifier (0.5–12% captured). **C3, fault observability:** a six-fault telemetry perturbation analysis, including three held-out fault classes, shows fault-specific over-flagging, discrimination, and active misclassification — a telemetry-only probe, without relevance labels, characterizes whether a new fault class is over-flagged, under-flagged, or discriminable. **C4, distribution shift:** a TAIL-calibrated verifier’s action-utility ranking becomes anti-correlated with true utility on HEAD/TORSO (AUC falling to 0.14–0.23). Two further findings are methodological lessons rather than headline contributions: an under-tuned BM25 baseline inflated every downstream comparison until corrected on validation alone (Section IV-C), and a real reranker-wiring defect, found and corrected mid-study, had been masking one fault’s recoverability (Section VII-A).

II. RELATED WORK

LLM query expansion. Query2doc [1] generates pseudo-documents from an LLM and appends them to the query; MuGI [2] generates and fuses multiple expansions. Both treat expansion as unconditional, one-shot generation, conditioning on neither retrieval-state telemetry nor a harmful rewrite’s recovery.

Agentic control and observability. ReAct [3] interleaves reasoning with actions and observations, the template for our structure, with a much smaller action space (typed retrieval operators, non-LLM controller). OpenTelemetry’s trace/metric/log separation and GenAI conventions [4] supply our telemetry’s naming discipline.

Query performance prediction and selective expansion. Classic query performance prediction [5] and selective query expansion [6] already ask whether a query is hard, or likely to benefit from expansion, from retrieval signal alone. We split that into two targets

sharing one telemetry vocabulary — is retrieval failing, versus will *this* action fix it — showing qrel-free telemetry answers the first far better (Section V).

Benchmarks. TripClick [7] has genuine head/torso/tail frequency structure, unlike benchmarks that treat “hard” as low-BM25-score — our reason for it as primary corpus.

III. SYSTEM DESIGN

A. Agent State and Control Loop

We use “agentic” operationally: a bounded system that observes state, selects actions, observes consequences, and adapts under a budget — not a claim of open-ended autonomy. The agent’s state at step t is $s_t = (q_0, q_t, R_t, O_t, H_t, B_t)$, where q_0 is the immutable original query, q_t the active search expression, R_t the current ranking, O_t the observability state, H_t the episode’s action/observation history, and B_t the remaining action budget. The control loop is OBSERVE $\rightarrow$ DIAGNOSE $\rightarrow$ ACT $\rightarrow$ RETRIEVE $\rightarrow$ OBSERVE $\rightarrow$ VERIFY $\rightarrow$ {ACCEPT, ROLLBACK, REPLAN, STOP}. The critical property this loop is built to test is that the second decision (accept/rollback/replan/stop) depends on evidence the agent’s own first action created — as opposed to a static pipeline that receives the same query, initial ranking, evidence, generator, and verifier, but executes exactly one predetermined route. Isolating this dependency, not simply comparing against unmodified BM25, is the paper’s central experimental contrast (Section VI).

The action space is deliberately small and interpretable: $A = \{ \text{NOEXPAND, VOCABULARY, CONTEXT, AMBIGUITY, ENTITYNORMALIZE, ROLLBACK, REPLAN, STOP} \}$. VOCABULARY maps informal user language onto corpus-compatible terminology; CONTEXT adds a corpus-supported missing facet to an underspecified query; AMBIGUITY disambiguates toward the interpretation the evidence best supports; ENTITYNORMALIZE resolves aliases, abbreviations, and canonical-name mismatches. The LLM is invoked through exactly one typed tool, `generate_expansion(action, query, evidence)`; it is never the sole arbiter of its own output’s use — that judgment is a separate, non-generative controller’s (Section III-D).

B. Observability Model

Every query episode produces one trace with spans `retrieve_original` $\rightarrow$ `build_evidence` $\rightarrow$ `plan_action` $\rightarrow$ `generate_expansion` $\rightarrow$ `retrieve_expanded` $\rightarrow$ `rerank` $\rightarrow$ `verify` $\rightarrow$ `recover_or_accept` $\rightarrow$ `fuse`. O_t spans three telemetry families. *Retrieval*: top score, score margin, normalized score entropy, query-term coverage, mean rare-term IDF, entity overlap, top- k ranking overlap against the previous state, reranker/BM25 disagreement, lexical result coherence, and ranking stability under a small controlled query perturbation. *Agent*: selected operator, trajectory length, previous verifier score, rewrite-to-original semantic drift, introduced entities, repeated actions, remaining budget. *System*: per-tool latency, token counts, tool errors/timeouts. Every feature is computable without relevance judgments, which is what makes the verifier usable at inference time.

C. Failure Taxonomy

We label episodes into ten failure categories (vocabulary mismatch, underspecification, ambiguity, entity mismatch, intent drift, confirmation bias, retriever disagreement, tool degradation, budget exhaustion, healthy retrieval), each with a telemetry signature and preferred recovery action, motivating Section VII’s fault-injection protocol.

D. Verifier and Controller

The verifier, `verify(previous_state, new_state)`, produces a scalar judgment of whether an action’s consequences look like an improvement, using telemetry alone. Rather than hand-picking its weights, we calibrate them against click-derived qrels (Section IV): execute one expansion action, compute the true nDCG@10 change against those labels, and fit the verifier’s weights to predict that change from Section III-B’s telemetry features. Click labels are available only during this offline calibration; the deployed verifier never sees one, and is treated as an artifact — trained by a script, loaded at inference time. Throughout this paper, “qrel-free” describes this deployment/inference regime specifically, not the method as a whole: the weights, thresholds, and action-utility targets that make inference qrel-free are themselves fit offline against click-derived qrels, and that dependency does not disappear at deployment — it resurfaces as a generalization question (Section VIII). The recovery controller is a small deterministic finite-state machine on the verifier’s scalar score v : $v \leq -0.102 \Rightarrow \text{ROLLBACK}$ to the pre-action ranking; $v \geq -0.081 \Rightarrow \text{ACCEPT}$, fusing the new ranking via reciprocal-rank fusion; otherwise REPLAN with an untried operator, capped at two model calls per query. Both cutoffs come from TAIL validation only (accept: median; harmful: 25th percentile), never test.

IV. EXPERIMENTAL SETUP

A. Datasets

TripClick [7] is the primary benchmark: 1,523,878 documents, split evenly across head/torso/tail frequency buckets, each further split into train/validation/test (1,175 queries per split per bucket). All results concern the TAIL split unless stated otherwise (Section VIII). Labels are click-derived, not human-assessed (binary raw clicks, matching the label type this split’s published reference point uses [7]); we use “qrel” and “click label” interchangeably, neither implying editorial judgment.

B. Experimental Protocol

A first pass at this study calibrated the verifier and thresholds on the same episodes later used to report RQ1–RQ3 — an in-sample characterization unsuited to an out-of-sample generalization claim. We froze a stricter protocol instead, in a committed config rather than left implicit: TAIL *validation* may be used for any retrieval-config, verifier, or threshold decision; TAIL *test* may be touched at most once per research question for its *reported* result, with no decision afterward conditioned on a test-set outcome. Implementation defects that break the described apparatus itself are a distinct case, and we hold them to a narrower standard than “never touched” rather than claim one that does not survive scrutiny: a defect is diagnosed because a result is self-evidently implausible (an ablation with no effect, a baseline far below a published reference), not because a plausible result was disliked, and no model, threshold, or feature choice is ever changed in response to what a diagnostic touch shows — only the broken component itself is fixed. Both defects found in this study (Sections IV-C, VII-A) were caught this way, and both required accessing TAIL test more than once as a direct, disclosed consequence: a reduced-scale diagnostic touch under a broken implementation, then one full-scale evaluation under the corrected pipeline. We report the complete access history for each, including what the pre-fix numbers were, rather than assert a single-touch claim the history does not support. This discipline directly produced the paper’s first finding, below.

C. A Corrected Retrieval Baseline

BM25 over an unstemmed FTS5 index scored nDCG@10 0.228 on TAIL test — 15% below the 0.267 reference for this exact split and label type [8]. Selecting entirely on validation, we found two config choices that close this gap: Porter stemming (val nDCG@10

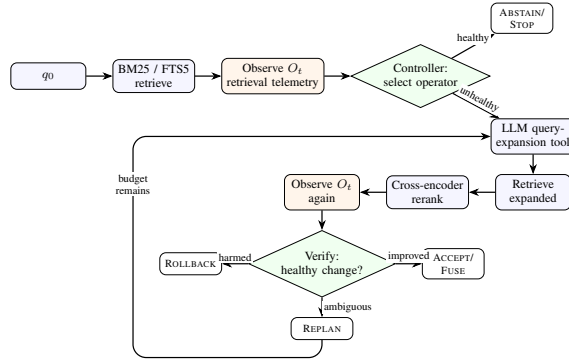


Fig. 1. The observability-guided control loop. Telemetry O_t (orange) drives both operator selection and the accept/rollback/replan/stop decision. The static comparator (Section VI) runs the same chain once, with no path back from VERIFY to the LLM tool.

0.235 $\rightarrow$ 0.270) and weighting the title field ten times body weight in a title/body-split FTS5 index (val nDCG@10 climbing to a clear plateau at that ratio, 0.327, MRR peaking there and declining beyond it). A single, one-time sanity check of the frozen config against withheld test — not a selection criterion — gives nDCG@10 0.315, above the published reference. We report this because an under-tuned baseline mechanically inflates the apparent value of anything compared against it: every downstream conclusion in this paper changed once this fix was in place, a precondition for the rest of the study rather than an implementation footnote.

D. Implementation

Sparse retrieval uses BM25 over a SQLite FTS5 index [9] rather than a JVM Lucene index, forced by a JVM crash under Pyserini [10] on our hardware; two further silent-corruption risks surfaced at scale, the same category as Section IV-C’s baseline gap and Section VII-A’s telemetry defect: a conjunctive default MATCH operator (collapsing recall without an error) and an unindexed companion lookup that degrades to a full table scan (18 $\times$ slower at 1.5M rows). Cross-encoder reranking (cross-encoder/ms-marco-MiniLM-L6-v2 [11]) applies to the top 100 BM25 candidates, reranking the top 50. Generation uses qwen2.5:3b (3.1B parameters, Q4_K_M 4-bit quantization) served locally via Ollama, greedy decoding (temperature 0), capped at two calls per query episode with a 4096-token context window and a 32–64 token structured output, so the language model is never the entire decision surface, only one typed tool within it. Prompts are corpus-grounded, one fixed template per expansion operator (src/expansion/prompts.py in the linked repository); telemetry feature definitions are listed in full in Section III-B. All experiments run on a single consumer machine (Apple M4 MacBook Air, 16GB unified memory) — no GPU cluster or cloud spend; per-experiment call counts and latency are reported where relevant (e.g. Section VI) rather than as an aggregate study-wide cost figure, which was not tracked. Every data split, bootstrap resample, and stochastic subsample in this paper uses the same fixed seed (42). Code, configs, and analysis scripts are available at an anonymized repository for review: <https://anonymous.4open.science/r/agentiq-query-expansion-8939/>.

E. Verifier Calibration Methodology

Following Section IV-B, we collect calibration episodes (one expansion action per query, true nDCG@10 change against click labels) on TAIL validation ($n = 1147$ – 1168). The calibration action comes from the same ActionSelector policy used everywhere else (Section III-D), not separate sampling, so its operator distribution matches the agent’s actual usage by construction. Operator identity is not itself a verifier feature (Section III-B): the fitted verifier estimates generic post-action utility, marginalized over whichever operator was selected, not utility conditional on a specific operator.

TABLE I
VALIDATION CHARACTERIZATION OF CANDIDATE VERIFIER FAMILIES ON TAIL validation ($n = 1147$ – 1168): PEARSON r VS. TRUE NDCG@10 CHANGE. NOT THE GENERALIZATION RESULT (TABLE II).

Model, feature family	Val Pearson r
v0 hand-picked weights (6 features)	+0.020 (n.s., $p=0.50$)
Linear, 10 features (lexical+reranker)	−0.014
GBDT, 10 features	+0.001
Linear, 13 features (+dense embedding)	−0.015

We compare three feature families of increasing size (six retrieval-telemetry deltas; plus four reranker/length features; plus three dense-embedding features) against three model classes each (fixed-weight, 5-fold CV linear, 5-fold CV GBDT), under Pearson correlation (Table I), to characterize the detection ceiling rather than accept the first plausible fit — never itself reported as evidence the signal generalizes (Section V does that separately, from a single evaluation on test under the frozen configuration, not repeated or selected among). Recovery thresholds are calibrated the same way, from candidate cutoffs on validation.

F. Metrics

Retrieval quality is nDCG@10, Recall@100, and MRR, plus the fraction of queries helped/unchanged/harmed relative to a baseline. For the recovery study (Section VII) we report Detection Recall $P(\text{intervened} \mid \text{harmful})$, False Alarm Rate $P(\text{intervened} \mid \text{healthy})$, and recovery gain in mean nDCG@10 with vs. without recovery on identically corrupted episodes. We additionally report a retrospective *oracle ceiling*: per query, $\max(\text{baseline}, \text{treatment})$ under true click labels, diagnosing whether headroom exists independent of whether our qrel-free verifier can find it (Section VI). Significance is via paired bootstrap (10,000 resamples) throughout.

V. RQ1: DETECTION

Nothing clears noise on validation: every candidate — a hand-picked baseline, linear and gradient-boosted fits on the deployed 10-feature set, and a fit augmenting it with dense-embedding features — lands at or below $|r| = 0.02$. We froze the 10-feature linear model before test evaluation: the only candidate fit on the corrected pipeline’s distribution rather than a stale or untested one (configs/experimental_protocol.yaml). GBDT-10, fit on the identical distribution, was statistically indistinguishable ($r=+0.001$ vs. -0.014) and was not substituted after seeing test results. We deploy the linear model *despite* its weak validation estimate, per Section IV-B: no decision is conditioned on test performance.

The frozen action-utility model generalizes better than its noisy validation estimate suggested: $r = 0.114$ on test is small but real ($p < 0.001$, $R^2 \approx 1.3\%$), modestly above no-skill baselines (Figure 2).

TABLE II

RQ1’S FROZEN RESULTS: BOTH MODELS EVALUATED ON WITHHELD TAIL TEST ($n = 1157$) UNDER THE FROZEN, CORRECTED CONFIGURATION; NO SUBSEQUENT MODEL SELECTION WAS CONDITIONED ON THESE RESULTS. BRACKETED VALUES ARE BOOTSTRAP 95% CIs (SCRIPTS/17).

Metric	Test value
Action utility: Pearson r vs. true Δ nDCG@10	0.114 ($p < 0.001$)
Action utility: ROC-AUC (detecting harm)	0.552 [0.518, 0.586]
Action utility: average precision (prev. 0.330)	0.364
Failure detection: ROC-AUC (val CV 0.618)	0.649 [0.618, 0.680]

TABLE III

RQ2, FROZEN RESULT, FULL TRIPCLICK TAIL TEST ($n = 1175$), CORRECTED BASELINE. STATIC QE AND AGENT SHARE RETRIEVER, GROUNDING, GENERATOR, VERIFIER, FUSION; AGENT ADDS A CLOSED OBSERVE-VERIFY-RECOVER LOOP.

System	nDCG@10	Recall@100	MRR
BM25 only	0.3152	0.7750	0.3014
BM25 + rerank, no QE	0.2850 [†]	0.7750	0.2701
BM25 + QE, no rerank	0.2992 [‡]	0.7716	0.2856
Static QE (QE + rerank)	0.3159	0.7759	0.2993
Failure-selective QE [§]	0.3179	0.7749	0.3019
Agent	0.3161	0.7770	0.3015

[†]: significantly below BM25 only, -0.0302 [$-0.0447, -0.0154$], $p < 0.001$. [‡]: significantly below BM25 only, -0.0160

[$-0.0237, -0.0086$], $p < 0.001$. QE $\times$ rerank interaction (paired bootstrap, scripts/13): $+0.0468$ [$+0.0303, +0.0630$], excludes zero. §: routes to Static QE iff pre-action failure probability exceeds a validation-median threshold, no post-action verifier or agent loop (scripts/31); gain vs. BM25 $+0.0027$ [$-0.0023, +0.0079$], not significant.

We read the val/test gap honestly rather than as vindication: cross-validation on $\sim 1,150$ episodes is a noisy estimator for an effect this small, and one larger test set landing positive does not retroactively make the selection sound.

A. Detection Is Not Actionability

That action-utility model answers a hard question: given telemetry before *and* after a specific action, did it help? A logically prior, cheaper question uses only pre-action telemetry: is unexpanded retrieval already unhealthy? We build a second model from the same eight retrieval-telemetry features, labeling a query “poor” when BM25 surfaces zero relevant documents in the top 10 (test prevalence 0.450), selected by 5-fold CV on validation and evaluated on frozen test under that fixed configuration, with no subsequent selection conditioned on the result, identically to Section IV-E’s protocol — Table II’s last row.

On frozen test, telemetry is moderately predictive of *whether* retrieval is failing (AUC = 0.649, 95% CI [0.618, 0.680]) but markedly weaker at predicting *whether* a *specific candidate action fixes it* (AUC = 0.552, 95% CI [0.518, 0.586]) — both evaluated once under the same frozen, corrected features and protocol, with no selection conditioned on either outcome, not the same result at different noise levels. Non-overlapping CIs alone don’t formally test the difference; a paired bootstrap on the $n=1157$ acted-upon queries (failure AUC on this subset: 0.643) gives $+0.0905$, 95% CI [$+0.0446, +0.1361$], excluding zero (scripts/28) — the gap itself is statistically established, not just the two point estimates. **RQ1’s answer:** qrel-free telemetry is moderately predictive of whether something is wrong, but even after observing the retrieval consequences of its own action, it is only weakly informative about whether that action actually helped. Both RQ2’s control result and RQ3’s fault-specific pattern trace back to this weakness.

VI. RQ2: CONTROL

Once BM25 is corrected (Section IV-C), the comparison changes: neither Static QE nor Agent significantly outperforms BM25 alone

TABLE IV

UNCONSTRAINED ORACLE CEILING, SAME EPISODES AS TABLE III. ORACLE PICKS $\max(\text{BM25}, \text{TREATMENT})$ PER QUERY, TRUE CLICK LABELS, NO BUDGET CAP.

Comparison	nDCG@10 gain	95% CI
Oracle-static vs. BM25	$+0.0246$	[0.0197, 0.0296]*
Oracle-static vs. Static QE	$+0.0239$	[0.0191, 0.0292]*
Oracle-agent vs. Agent	$+0.0276$	[0.0226, 0.0332]*

*: significant, $p < 0.001$.

(static: $+0.0007$ nDCG@10, 95% CI [$-0.0066, 0.0079$]; agent: $+0.0009$, 95% CI [$-0.0069, 0.0087$]), and Agent does not significantly outperform Static QE ($+0.0002$, 95% CI [$-0.0027, 0.0032$]; changes 3.9% of outcomes, 2.1% helped, 1.7% harmed). The agent still costs 24% more model calls (1.22 vs. 0.98 per query) and 36% higher mean latency (15.1 s vs. 11.1 s, excluding sleep-inflated episodes, isolated per-pipeline batches with alternated run order and a discarded warm-up call to avoid cache and drift confounds, scripts/07) for no offsetting gain. Decomposing the two components (Table III) as a 2×2 factorial and computing the interaction directly, $I = \text{mean}(\text{QE} + \text{rerank}) - \text{mean}(\text{QE}) - \text{mean}(\text{rerank}) + \text{mean}(\text{BM25})$, gives $I = +0.0468$, 95% CI [$+0.0303, +0.0630$], excluding zero: a genuine interaction, not one component offsetting a fixed cost from the other. Reranking alone is significantly *harmful* (-0.0302 , $p < 0.001$) and expansion alone is *also* significantly harmful (-0.0160 , 95% CI [$-0.0237, -0.0086$]) — neither component helps in isolation. Yet adding the reranker back once expansion is already applied significantly *helps* ($+0.0167$, 95% CI [$0.0076, 0.0257$]), the opposite sign from adding it to raw BM25: the cross-encoder, trained on general web relevance, misorders raw BM25’s candidate pool but behaves better once expansion has already reshaped it. Static QE’s near-parity with BM25 is this interaction landing close to neutral, not one component compensating for the other. **On natural queries, neither expansion flavor moves the needle once the retriever is competitive** — a weaker baseline would have made both look like they help.

RQ1’s finding suggests a cheaper test: if failure detection (AUC 0.649) is the strong half of the signal, does routing on it *alone* — no verifier, no rollback, no agent loop, just Static QE when the pre-action failure probability exceeds a validation-chosen threshold and BM25 otherwise — do any better than either unconditional policy? We re-combine already-frozen per-query outcomes under this new, pre-specified rule (never swept to flatter the comparison; scripts/31) and add it to Table III. Its point estimate is the best of the three real policies ($+0.0027$ nDCG@10 vs. BM25, routing 48.7% of queries to QE) — higher than unconditional Static QE ($+0.0007$) and higher than the full closed-loop Agent ($+0.0009$) — but, like both, not significantly different from BM25 (95% CI [$-0.0023, +0.0079$]), nor significantly different from either of them directly (vs. Static QE: $+0.0020$, 95% CI [$-0.0030, +0.0072$]; vs. Agent: $+0.0018$, 95% CI [$-0.0041, +0.0078$]). We read this as mildly favorable to the failure-selective controller without treating it as established: detection alone gets no further than unconditional expansion or the full agent do, on this sample size, even though it uses the paper’s single strongest signal and none of its weakest one.

A. Is There Any Headroom at All?

A null result here is ambiguous: either no control opportunity exists, or one exists but this verifier cannot find it. We resolve this with a retrospective oracle: per query, take whichever of BM25 or the treatment ranking true click labels say is better — not deployable, but useful to bound the second explanation.

The oracle shows substantial headroom: a perfect selective controller gains $+0.0246$ nDCG@10 over BM25 ($p < 0.001$), whereas unconditional Static QE gains $+0.0007$ and is not significantly different from BM25. Static expansion helps 11.6% of queries and harms 10.6% relative to BM25, a real, roughly-balanced mix a

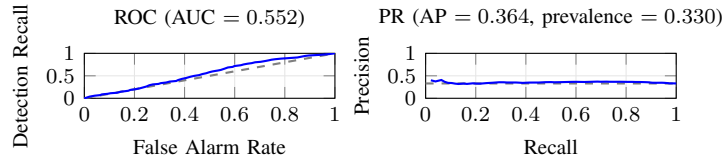


Fig. 2. RQ1’s frozen result: ROC and precision-recall curves for the selected verifier’s score against true harm (nDCG@10 $\Delta < 0$) on withheld TAIL test ($n = 1157$), evaluated once under the frozen configuration with no subsequent selection conditioned on the result. The curve sits consistently, but only modestly, above the no-skill baseline (dashed) in both panels.

TABLE V

SELECTIVITY FRONTIER, TAIL TEST ($n = 1175$, `SCRIPTS/14`). EACH POLICY TREATS *exactly* $X\%$ OF QUERIES WITH STATIC QE (AN INTERVENTION QUOTA, NOT A CAP), RANKED BY TRUE BENEFIT (ORACLE) OR VERIFIER SCORE (VERIFIER), REST AT BM25; MEAN NDCG@10 GAIN OVER ALWAYS-BM25. ORACLE GAIN NEED NOT BE MONOTONE IN X : PAST THE FRACTION TREATMENT TRULY HELPS, IT IS FORCED ONTO HARMFUL QUERIES TOO. % COLUMN: VERIFIER GAIN AS A FRACTION OF ORACLE GAIN.

Quota	Verifier gain	Oracle gain	%
5%	0.0001	0.0184	0.5%
10%	0.0006	0.0242	2.5%
20%	0.0025	0.0246	10.2%
40%	0.0029	0.0246	11.8%
100%	0.0007	0.0007	100%*

*At 100% quota every policy treats every query, so ranking cannot matter; verifier and oracle necessarily coincide with Table III’s unconditional Static QE result.

perfect selector could exploit. The evidence points to *verification* as the dominant bottleneck: RQ1’s action-utility verifier, $r = 0.114$ on frozen test, is far too weak to discriminate the two groups — converting RQ2’s null result from “maybe there’s nothing here” into “there is something here, and our qrel-free verifier cannot see it.”

This oracle is susceptible to a specific optimism: it selects the better system separately for every query using the same sparse, noisy click labels the gain is then evaluated on, which can inflate the reported headroom beyond what a real, imperfect selector could ever capture. We check this directly rather than assert it away, by computing the identical oracle definition on TAIL *validation* (never used to select the oracle itself, only to calibrate the verifier and BM25 config) as an out-of-sample stability check (`scripts/30`). The two disagree sharply: validation gives $+0.0533$ nDCG@10, 95% CI $[+0.0452, +0.0617]$ ($n=1147$) — more than double the frozen test figure above, with non-overlapping CIs. On the NQ replication, by contrast, the same comparison is far more stable: validation oracle $+0.0399$ vs. test/extension oracle $+0.0415$ (Section IX). We read this as evidence that the oracle magnitude is substantially split-sensitive on TripClick — unlike NQ, where the two nearly agree, suggesting the instability is tied to TripClick’s sparser per-query click signal rather than the oracle definition itself. This discrepancy does not by itself show which estimate, if either, is closer to a real selector’s reachable gain: both use per-query maximization against the same kind of sparse click labels they are evaluated on, so split disagreement demonstrates instability, not which direction the bias runs. We therefore interpret $+0.0246$ as directional evidence that heterogeneous per-query treatment effects exist, rather than as a precise estimate of deployable selective-expansion headroom. A fully debiased estimator (cross-fitting the selection and evaluation folds, or averaging over repeated relevance draws) would characterize this more precisely than a single cross-split comparison can, and is left for future work.

Oracle gain is flat at 0.0246 from 20% to 40% (both quotas still fall within the 11.6% helped plus 77.8% unchanged, contributing zero either way) before collapsing to $+0.0007$ at 100%, once forced to also treat the remaining 10.6% genuinely harmful queries — exactly

TABLE VI

AGENT CONTROLLER-DECISION FREQUENCY, FULL TRIPCLICK TAIL TEST SET ($n = 1175$).

Decision	Count	%
ACCEPT	699	59.5
ROLLBACK	331	28.2
REPLAN	127	10.8
ABSTAIN (no expansion)	18	1.5

TABLE VII

COMPLETE TAIL-TEST ACCESS HISTORY FOR RQ3. “AFFECTED FINAL NUMBERS?” MEANS WHETHER THAT STAGE’S OWN OUTCOME CHANGED ANY MODEL, THRESHOLD, OR ANALYTICAL CHOICE — NOT WHETHER A LATER, PURELY MECHANICAL RE-EVALUATION SUPERSEDED IT.

Stage	Split	Affected a choice?
Pilot fault-injection run ($n=150$ /fault type)	TAIL test, reduced scale	No — surfaced an implausible result (recall 0.185 for an ablation that should degrade something), which triggered code inspection, not re-analysis
Reranker-wiring fix	implementation only	N/A: a code correction, not a data-conditioned decision
BM25 baseline correction	TAIL validation (selection) + one TAIL test sanity check	Yes, but pre-committed: config chosen entirely on validation; test used once to confirm, not to select (Section IV-C)
Full-scale replication ($n=1175$)	TAIL test	Yes — this is the reported result; no further tuning followed

what a true quota predicts. At every quota below universal treatment, the verifier captures only a small fraction of available headroom, 0.5–11.8%, never close to the oracle.

Table VI shows a controller that intervenes often (38.9% ROLLBACK or REPLAN) without those interventions paying off: the frozen thresholds are wide, since nothing on validation cleanly separated harmful from healthy episodes, so the controller second-guesses frequently but not accurately. **RQ2’s answer is no**, with the oracle clarifying *why*: an observability-guided closed loop built on this verifier does not outperform a matched static pipeline, not because no opportunity exists, but because this verifier cannot exploit it.

VII. RQ3: WHICH FAILURES ARE RECOVERABLE?

RQ1 and RQ2 leave open whether observability has *any* practical value, or whether a weak general detector is useless. We test a narrower claim: that the same weak verifier can serve as a safety net against *large, obvious* failures even though it cannot reliably judge *subtle, natural* ones. We inject three fault types at their natural point of corruption — an unsupported entity appended to the expansion, a grounding passage swapped for a non-supporting one, and the reranker disabled — and compare, on identical corrupted episodes, an always-accept ablation against the real controller.

A. A Defect That Changed the Answer

TAIL test was accessed more than once for this research question. We report the complete sequence rather than a single-touch claim:

Concretely: an early pilot ($n=150$ per fault type, TAIL test) found the reranker-disabled fault essentially undetectable (recall 0.185) while the other two looked strong (entity injection recall 1.000, gain

TABLE VIII

RQ3, FROZEN RESULT: FAULT INJECTION, TAIL TEST ($n = 1157$). GAIN IS MEAN NDCG@10 WITH MINUS WITHOUT RECOVERY ON IDENTICAL EPISODES; CI FROM 10,000-RESAMPLE BOOTSTRAP. DOMINANT CONTRIBUTOR: FEATURE j WITH LARGEST $|w_j \Delta x_j|$.

Fault type	Rec.	FA	Gain (95% CI)	Dominant contrib.
Entity injection	0.974	0.939	+0.0081 [-0.003, +0.019]	coverage_gain -0.054
Grounding corruption	0.528	0.469	+0.0099 [+0.002, +0.018]*	topk_overlap +0.002
Reranker disabled	0.000	0.002	+0.0000	disagreement +0.126

CIs straddle zero except *: significant, positive. FA: false alarm.

+0.0238; grounding corruption recall 0.773, gain +0.0278 — both since superseded). An ablation that disables a pipeline stage showing no effect at all is implausible on its face, independent of whether one likes or dislikes the number; tracing it revealed a real defect: the reranked ranking was computed, at real latency cost, but never actually applied to the accepted ranking — disabling the reranker was mechanically a no-op regardless of what the fault-injection code did. Fixing it exposed a second, entangled defect: the verifier’s telemetry was still computed from the pre-rerank ranking. We corrected both, re-ran calibration on TAIL *validation* to confirm the deployed verifier did not need retraining (corrected Pearson $r = 0.146$ vs. 0.134), and then ran the full $n=1175$ TAIL *test* replication reported in Table VIII, with Section IV-C’s baseline correction — a shared upstream component — also already applied; that correction is the other reason the reported numbers differ from the pilot. No model, threshold, or feature choice was ever changed because of what the pilot or the pre-BM25-fix run showed — both triggered fixes to the apparatus itself, and the table above discloses every access rather than compressing it into “single touch.” We report this because a component computed purely for observability is architecturally easy to leave disconnected from the outcome it explains, and the error looked coherent — unrealistically strong — until an implausible near-zero effect prompted investigation.

B. Frozen Results

The three fault types (Table VIII) split into three distinct outcomes: no single answer to “is recovery useful,” only a per-failure-type one. **Entity injection** is detected almost every time but flagged almost as often, netting out non-significant. **Grounding corruption** is the one clean recovery success. **Reranker disabled** is never detected: the controller essentially always accepts.

Decomposing the score shift. Fitted weights alone show which features the verifier ignores, not what a fault *does* to its score: $\Delta x_j = \mathbb{E}[x_j^{\text{fault}} - x_j^{\text{clean}}]$, $\text{Contribution}_j = w_j \cdot \Delta x_j$ ($n=79/\text{fault}$) — a decomposition against the verifier’s fitted (co-varying) features, not an independent causal effect of feature j alone. The clearest case is reranker disabled: its dominant contribution is large and *positive*, from disagreement moving in the direction the verifier reads as improvement — it does not merely evade weak reranker-derived features, it makes the corrupted state score higher, so the controller is actively misled, not merely blind.

RQ3’s answer is that recoverability is fault-specific: a real net gain needs a score shift small enough for per-episode discrimination (grounding), is a wash when the shift is large and uniform (entity injection), and a fault is disguised as an improvement when it moves a strongly-weighted feature the *wrong* way (reranker) — a verifier can be actively adversarial to a fault it was never calibrated against.

C. A Diagnostic Rule, Tested on Three Held-Out Faults

The three faults’ net contributions ($\sum_j w_j \Delta x_j$, Table IX) and recall/false-alarm pairs move together: strongly negative pushes the score down regardless of true harm, giving high recall but comparably high false alarms (entity injection); strongly positive does the opposite (reranker disabled); only near-zero leaves room for per-episode variation to separate harmful from healthy cases — the one fault with a significant net gain (grounding). This needs only paired telemetry,

TABLE IX

NET CONTRIBUTION VS. RECALL, SORTED. GAIN: FROZEN SIGNIFICANCE FOR THE ORIGINAL THREE ($n=1157$); HELD-OUT ROWS ($n=79-80$) ARE POINT ESTIMATES.

Fault	Net contrib.	Recall	FA	Gain
Entity injection	-0.0529	0.974	0.939	NS
Query-term removal†	-0.0236	0.800	0.694	+0.0012
Grounding corruption	+0.0002	0.528	0.469	*
Append irrelevant term†	+0.0157	0.235	0.258	-0.0185
Truncated pool†	+0.1225	0.000	0.000	+0.0000
Reranker disabled	+0.1296	0.000	0.002	NS

†: held out from rule derivation; regime characterized from the fault’s telemetry shift before evaluating recovery quality (post-probe, not zero-shot). *: significant, $p < 0.05$. NS: not significant.

TABLE X

CROSS-SPLIT ROBUSTNESS. BM25/STATIC QE/ORACLE GAIN: FULL TEST SPLITS ($n = 1175$). HEAD/TORSO ACTION-UTILITY AUC: BOOTSTRAP 95% CI ON A FIXED, RANDOMLY SAMPLED $n = 400$ SUBSET PER SPLIT (SEED 42, `SCRIPTS/16`).

Split	BM25	Static QE	Oracle gain	AUC [95% CI]
HEAD	0.2282	0.2293 (NS)	+0.0153*	0.143 [0.107, 0.180]
TORSO	0.2485	0.2506 (NS)	+0.0187*	0.234 [0.189, 0.281]
TAIL	0.3152	0.3159 (NS)	+0.0246*	0.552 [0.518, 0.586]

NS: CI crosses zero. *: significant, $p < 0.001$.

no relevance labels: it is usable after a telemetry-only probe of a new fault, without labels or a full recovery-quality evaluation.

We evaluated this on three held-out faults never used to derive the rule (`scripts/12`, 18, 19): removing the query’s most discriminative term, appending an irrelevant term, and truncating the retrieval pool. **Human intuition about a new fault went 1-for-3:** correct for query-term removal, but wrong for append-irrelevant-term (guessed it would pattern-match; it came out weakly positive, low recall) and wrong for truncated pool (guessed grounding-like; it came out strongly positive, recall 0.000, same dominant feature as reranker disabled). Across the six tested fault classes, sorting by net telemetry contribution reproduced the observed recall ordering (Table IX). For the three held-out faults, this should be read as post-probe characterization rather than zero-shot prediction (Section XI): the qrel-free statistic, needing no guessing, tracked recoverability regime across content, evidence, system-component, and query-level corruption alike where our own guesses often did not. Whether that regime converts into a significant gain remains harder: only grounding produced one.

VIII. DOES THE VERIFIER GENERALIZE BEYOND TAIL?

Every calibration decision so far used TAIL validation only; we did not test whether the frozen verifier’s ranking holds on HEAD/TORSO, which it never saw during selection. Using the identical config, no recalibration, we run BM25 and Static QE on each bucket’s test split.

Static QE stays statistically indistinguishable from BM25 on every bucket, matching TAIL; oracle headroom exists on every bucket too. What does not transfer is the verifier: action-utility AUC falls from 0.552 on TAIL (95% CI [0.518, 0.586]) to 0.234 on TORSO ([0.189, 0.281]) and 0.143 on HEAD ([0.107, 0.180]) — both CIs entirely *below* chance, anti-correlated with true utility, not merely uninformative. A verifier calibrated on one frequency regime does not just fail to add value in another; its ranking inverts. This is the cost of Section III-D’s qrel-dependent calibration made concrete: the weights are an artifact of TAIL’s click-label distribution specifically, not a domain-general signal, and qrel-free *inference* does nothing to protect against that regime mismatch. These HEAD/TORSO estimates use fixed $n=400$ subsets for comparability across splits (unlike TAIL’s full $n=1175$), so read them as evidence of ranking inversion rather than precise bucket-level performance figures.

TABLE XI

TRIPCLICK TAIL VS. BEIR NQ [12], [13] ($n=2,052$ NQ CONFIRMATORY QUERIES, EVALUATED ONCE WITH NO DECISION CONDITIONED ON THE OUTCOME; PILOT/PROTOCOL DETAILS IN SECTION IX). BOOTSTRAP 95% CIs. NS: CROSSES ZERO. *: $p < 0.001$.

Corpus	Static QE gain	Oracle gain	Failure AUC	Action-util. AUC
TripClick	+0.0007 (NS)	+0.0246*	0.649 [.618, .680]	0.552 [.518, .586]
BEIR NQ	+0.0025 (NS)	+0.0415*	0.680 [.658, .703]	0.556 [.517, .594]

IX. DOES THE FINDING REPLICATE ON A SECOND CORPUS?

To check whether Section V-A’s finding is specific to TripClick, we replicate the detection-vs-actionability comparison on BEIR Natural Questions [12], [13], an open-domain corpus unrelated to TripClick’s clinical queries. NQ ships as one 3,452-query split with no train/val/test partition, so we carve a disjoint, fixed-seed (42) val/test split (800/600), freeze a fresh BM25 config and detectors on val only (same features/model classes as TripClick’s, refit since corpus-specific). An initial $n=600$ pilot touch of test — analogous to Section IV-B’s own TripClick pilot — was directionally consistent but too imprecise to resolve the ordering (action-utility AUC CI straddled chance). We therefore ran the same frozen pipeline, unchanged, exactly once as a confirmatory test on the entire remaining, disjoint pool of 2,052 never-before-touched NQ queries, with no decision conditioned on the outcome, reported in full regardless of what it showed; the $n=600$ pilot is not pooled into it. Table XI reports this $n=2,052$ result: as on TripClick, Static QE does not significantly beat BM25 despite oracle-capturable headroom ($p < 0.001$), and the same ordering replicates on the disjoint confirmatory set — pre-action telemetry is more predictive of retrieval failure (AUC 0.680) than post-action telemetry is of intervention utility (AUC 0.556, 95% CI [0.517, 0.594], clearly above chance). A paired bootstrap of the AUC difference on the $n=2,013$ acted-upon queries excludes zero (+0.119, 95% CI [+0.076, +0.162], *scripts/29*). This is an independent second-corpus replication of Section V-A’s ordering, not evidence the finding generalizes broadly across domains — one further corpus, one disjoint 2,052-query touch.

X. DISCUSSION

RQ1–RQ3, the oracle analysis, the diagnostic rule, and the cross-split check sketch one finding. Section V-A is load-bearing: telemetry is moderately predictive of whether retrieval is failing (AUC 0.649) but weakly informative about whether a specific action on an ordinary query will fix it (AUC 0.552). RQ2 shows the consequence: a closed loop built on the weaker signal adds nothing over a matched static pipeline, and the oracle/selectivity-frontier analyses indicate +0.0246 nDCG@10 of directional headroom (Section VI-A) that the verifier reaches at most 12% of. RQ3’s attribution shows this is a per-fault property, sharp enough in one case that the verifier reads the fault as an improvement. The diagnostic rule (Section VII-C) provides the clearest positive evidence: for *large, structured* faults, the same weights characterize which regime a new fault falls into after a telemetry-only probe. Section VIII bounds this differently, holding the fault fixed and varying query frequency instead: the diagnostic rule transfers to held-out fault types after a telemetry-only probe, whereas the action-utility ranking does not transfer across query-frequency regimes.

Both the BM25 baseline and reranker-wiring defects looked coherent rather than broken: a frozen protocol with a mandatory sanity check caught the first, distrusting a too-good result caught the second.

XI. LIMITATIONS

The oracle ceiling (Section VI-A) is retrospective, using test-set click labels directly — a valid upper bound, never deployable, and, on TripClick specifically, unstable across splits (validation oracle more than double the test figure, *scripts/30*); read it as directional

evidence of headroom, not a tight bound. The generator is a locally hosted, quantized 2–3B parameter model under a two-call budget; a larger generator might change RQ1’s ceiling or RQ2’s null result. Cross-split robustness covers TripClick’s three buckets, and the core detection-vs-actionability finding replicates on one further corpus (Section IX); broader coverage remains untested. Click-derived qrels are an imperfect relevance proxy inherited from TripClick. The diagnostic rule (Section VII-C) orders all six fault types’ detection recall correctly, though our own guesses about each fault’s regime were often wrong; six fault types is still small and hand-designed, and post-probe rather than zero-shot: it needs a fault’s own telemetry shift to compute. The failure-detection model uses one binary definition of “poor retrieval”; a graded definition could shift its measured AUC.

XII. CONCLUSION

We evaluated a closed-loop, observability-guided query-expansion agent under a corrected protocol separating model selection from final evaluation — surfacing the paper’s first finding: a validation-selected BM25 correction materially changed the downstream comparisons. Runtime observability is moderately predictive of retrieval failure (detection AUC 0.649), but autonomous recovery needs a stronger property the same telemetry does not deliver: predicting whether a specific intervention will help (action-utility AUC 0.552, below chance outside its calibration split). That gap is why the closed-loop agent does not outperform static expansion and neither outperforms BM25, despite a retrospective oracle indicating heterogeneous per-query headroom (+0.0246 nDCG@10 on TAIL test, Section VI-A). The recovery study shows this is a per-fault property with distinct, interpretable telemetry signatures — in one case the verifier is actively misled, not merely blind — and, tested on three held-out faults, supports label-free characterization of a new failure’s detection regime after a telemetry-only probe; converting that into a significant gain still needs a full evaluation. The same ranking inverts across query-frequency regimes, yet the core detection-vs-actionability gap itself replicates on a second, unrelated corpus (BEIR NQ, Section IX). Both implementation defects reported here (Sections IV-C, VII-A) also illustrate how retrieval and observability failures can produce coherent but misleading experimental conclusions.

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